TABLE CREATIONS

```
mysql> use aditya;
Database changed
mysql> CREATE TABLE Customers (
    ->     customer_id INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     age INT,
    ->     gender VARCHAR(10),
    ->     income DECIMAL(10,2),
    ->     employment_status VARCHAR(50),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Loans (
    ->     loan_id INT PRIMARY KEY,
    ->     customer_id INT,
    ->     loan_amount DECIMAL(12,2),
    ->     loan_type VARCHAR(50),
    ->     interest_rate DECIMAL(5,2),
    ->     loan_term INT,  -- in months
    ->     loan_status VARCHAR(20),  -- Approved / Rejected / Defaulted / Closed
    ->     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Credit_History (
    ->     credit_id INT PRIMARY KEY,
    ->     customer_id INT,
```

```
mysql> CREATE TABLE Credit_History (
    ->     credit_id INT PRIMARY KEY,
    ->     customer_id INT,
    ->     credit_score INT,
    ->     past_defaults INT,
    ->     credit_utilization DECIMAL(5,2),
    ->     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Payments (
    ->     payment_id INT PRIMARY KEY,
    ->     loan_id INT,
    ->     payment_date DATE,
    ->     amount_paid DECIMAL(10,2),
    ->     payment_status VARCHAR(20), -- On Time / Late
    ->     FOREIGN KEY (loan_id) REFERENCES Loans(loan_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Customers (customer_id, name, age, gender, income, employment_status, city) VALUES
    -> (1,'Customer_1',25,'Male',35000,'Salaried','Delhi'),
    -> (2,'Customer_2',32,'Female',48000,'Self-Employed','Mumbai'),
    -> (3,'Customer_3',45,'Male',72000,'Business','Bangalore'),
    -> (4,'Customer_4',29,'Female',29000,'Salaried','Lucknow'),
    -> (5,'Customer_5',38,'Male',85000,'Business','Hyderabad'),
    -> (6,'Customer_6',41,'Female',54000,'Salaried','Pune'),
    -> (7,'Customer_7',27,'Male',31000,'Salaried','Chennai'),
    -> (8,'Customer_8',50,'Female',90000,'Business','Kolkata'),
    -> (9,'Customer_9',34,'Male',60000,'Self-Employed','Jaipur'),
    -> (10,'Customer_10',28,'Female',45000,'Salaried','Delhi'),
    -> (11,'Customer_11',39,'Male',78000,'Business','Mumbai'),
    -> (12,'Customer_12',31,'Female',52000,'Salaried','Lucknow'),
    -> (13,'Customer_13',44,'Male',66000,'Self-Employed','Pune'),
    -> (14,'Customer_14',26,'Female',33000,'Salaried','Chennai'),
    -> (15,'Customer_15',36,'Male',58000,'Business','Hyderabad'),
    -> (16,'Customer_16',48,'Female',88000,'Business','Delhi'),
    -> (17,'Customer_17',30,'Male',47000,'Salaried','Bangalore'),
    -> (18,'Customer_18',42,'Female',69000,'Self-Employed','Kolkata'),
    -> (19,'Customer_19',37,'Male',61000,'Salaried','Mumbai'),
    -> (20,'Customer_20',29,'Female',42000,'Salaried','Lucknow');
Query OK, 20 rows affected (0.02 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

DATA INSERTION

```
mysql> INSERT INTO Customers (customer_id, name, age, gender, income, employment_status, city)
    -> SELECT
    ->     n,
    ->     CONCAT('Customer_', n),
    ->     FLOOR(25 + (RAND()*30)),
    ->     IF(RAND()>0.5,'Male','Female'),
    ->     FLOOR(25000 + (RAND()*75000)),
    ->     ELT(FLOOR(1 + (RAND()*3)),'Salaried','Business','Self-Employed'),
    ->     ELT(FLOOR(1 + (RAND()*6)),'Delhi','Mumbai','Bangalore','Lucknow','Pune','Hyderabad')
    -> FROM (
    ->     SELECT 21 n UNION SELECT 22 UNION SELECT 23 UNION SELECT 24 UNION SELECT 25
    ->     UNION SELECT 26 UNION SELECT 27 UNION SELECT 28 UNION SELECT 29 UNION SELECT 30
    ->     UNION SELECT 31 UNION SELECT 32 UNION SELECT 33 UNION SELECT 34 UNION SELECT 35
    ->     UNION SELECT 36 UNION SELECT 37 UNION SELECT 38 UNION SELECT 39 UNION SELECT 40
    ->     UNION SELECT 41 UNION SELECT 42 UNION SELECT 43 UNION SELECT 44 UNION SELECT 45
    ->     UNION SELECT 46 UNION SELECT 47 UNION SELECT 48 UNION SELECT 49 UNION SELECT 50
    ->     UNION SELECT 51 UNION SELECT 52 UNION SELECT 53 UNION SELECT 54 UNION SELECT 55
    ->     UNION SELECT 56 UNION SELECT 57 UNION SELECT 58 UNION SELECT 59 UNION SELECT 60
    ->     UNION SELECT 61 UNION SELECT 62 UNION SELECT 63 UNION SELECT 64 UNION SELECT 65
    ->     UNION SELECT 66 UNION SELECT 67 UNION SELECT 68 UNION SELECT 69 UNION SELECT 70
    ->     UNION SELECT 71 UNION SELECT 72 UNION SELECT 73 UNION SELECT 74 UNION SELECT 75
    ->     UNION SELECT 76 UNION SELECT 77 UNION SELECT 78 UNION SELECT 79 UNION SELECT 80
    ->     UNION SELECT 81 UNION SELECT 82 UNION SELECT 83 UNION SELECT 84 UNION SELECT 85
    ->     UNION SELECT 86 UNION SELECT 87 UNION SELECT 88 UNION SELECT 89 UNION SELECT 90
    ->     UNION SELECT 91 UNION SELECT 92 UNION SELECT 93 UNION SELECT 94 UNION SELECT 95
    ->     UNION SELECT 96 UNION SELECT 97 UNION SELECT 98 UNION SELECT 99 UNION SELECT 100
    -> ) numbers;
Query OK, 80 rows affected (0.02 sec)
Records: 80  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Credit_History (credit_id, customer_id, credit_score, past_defaults, credit_utilization)
    -> SELECT
    ->     customer_id,
    ->     customer_id,
    ->     FLOOR(450 + (RAND()*400)),  -- Credit score between 450-850
    ->     FLOOR(RAND()*3),
    ->     ROUND(20 + (RAND()*70),2)
    -> FROM Customers;
Query OK, 100 rows affected (0.01 sec)
Records: 100  Duplicates: 0  Warnings: 0
```

```
ysql> INSERT INTO Credit_History (credit_id, customer_id, credit_score, past_defaults, credit_utilization)
   -> SELECT
   ->     customer_id,
   ->     customer_id,
   ->     FLOOR(450 + (RAND()*400)),  -- Credit score between 450-850
   ->     FLOOR(RAND()*3),
   ->     ROUND(20 + (RAND()*70),2)
   -> FROM Customers;
uery OK, 100 rows affected (0.01 sec)
ecords: 100  Duplicates: 0  Warnings: 0

ysql> INSERT INTO Loans (loan_id, customer_id, loan_amount, loan_type, interest_rate, loan_term, loan_status)
   -> SELECT
   ->     n,
   ->     FLOOR(1 + (RAND()*100)),
   ->     FLOOR(100000 + (RAND()*900000)),
   ->     ELT(FLOOR(1 + (RAND()*3)),'Home Loan','Personal Loan','Car Loan'),
   ->     ROUND(7 + (RAND()*8),2),
   ->     ELT(FLOOR(1 + (RAND()*3)),12,24,36),
   ->     ELT(FLOOR(1 + (RAND()*4)),'Approved','Rejected','Closed','Defaulted')
   -> FROM (
   ->     SELECT 1 n UNION SELECT 2 UNION SELECT 3 UNION SELECT 4 UNION SELECT 5
   ->     UNION SELECT 6 UNION SELECT 7 UNION SELECT 8 UNION SELECT 9 UNION SELECT 10
   ->     UNION SELECT 11 UNION SELECT 12 UNION SELECT 13 UNION SELECT 14 UNION SELECT 15
   ->     UNION SELECT 16 UNION SELECT 17 UNION SELECT 18 UNION SELECT 19 UNION SELECT 20
   ->     UNION SELECT 21 UNION SELECT 22 UNION SELECT 23 UNION SELECT 24 UNION SELECT 25
   ->     UNION SELECT 26 UNION SELECT 27 UNION SELECT 28 UNION SELECT 29 UNION SELECT 30
   ->     UNION SELECT 31 UNION SELECT 32 UNION SELECT 33 UNION SELECT 34 UNION SELECT 35
   ->     UNION SELECT 36 UNION SELECT 37 UNION SELECT 38 UNION SELECT 39 UNION SELECT 40
   ->     UNION SELECT 41 UNION SELECT 42 UNION SELECT 43 UNION SELECT 44 UNION SELECT 45
   ->     UNION SELECT 46 UNION SELECT 47 UNION SELECT 48 UNION SELECT 49 UNION SELECT 50
   -> ) numbers;
uery OK, 50 rows affected (0.01 sec)
ecords: 50  Duplicates: 0  Warnings: 0
```

TABLE VIEW

```
mysql> select *from Loans;
```

loan_id	customer_id	loan_amount	loan_type	interest_rate	loan_term	loan_status
1	6	791847.00	Car Loan	7.29	12	Defaulted
2	33	367729.00	Personal Loan	12.80	12	Approved
3	34	454457.00	Car Loan	11.59	12	Rejected
4	71	511761.00	Home Loan	11.47	12	Rejected
5	75	342072.00	Home Loan	12.67	12	Approved
6	50	415738.00	Home Loan	9.57	36	Defaulted
7	42	308473.00	Car Loan	13.52	24	Rejected
8	89	301495.00	Personal Loan	12.39	36	Defaulted
9	6	911286.00	Personal Loan	7.30	12	Closed
10	62	334728.00	Personal Loan	10.67	36	Rejected
11	2	956400.00	Car Loan	12.80	24	Approved
12	62	500797.00	Personal Loan	11.65	36	Approved
13	90	443293.00	Home Loan	14.66	12	Defaulted
14	46	959199.00	Personal Loan	8.87	36	Defaulted
15	57	319528.00	Personal Loan	14.11	36	Closed
16	81	973284.00	Personal Loan	9.36	12	Defaulted
17	89	824478.00	Personal Loan	10.99	24	Approved
18	15	114523.00	Personal Loan	8.37	36	Approved
19	79	619182.00	Personal Loan	14.65	12	Defaulted
20	40	144793.00	Home Loan	7.86	24	Closed
21	82	376900.00	Home Loan	11.04	12	Defaulted
22	34	256302.00	Car Loan	13.83	24	Closed
23	36	832953.00	Car Loan	11.43	36	Approved
24	62	578105.00	Car Loan	10.44	36	Rejected
25	73	534936.00	Home Loan	12.59	36	Defaulted
26	27	566128.00	Car Loan	10.23	24	Approved
27	9	486068.00	Car Loan	7.98	36	Rejected
28	41	679833.00	Home Loan	7.79	24	Approved
29	1	779879.00	Car Loan	11.58	24	Approved
30	70	315926.00	Home Loan	14.20	12	Defaulted
31	34	839203.00	Home Loan	7.93	12	Defaulted
32	35	375337.00	Personal Loan	11.34	12	Closed
33	8	713976.00	Home Loan	14.66	12	Approved
34	67	270890.00	Car Loan	8.82	12	Closed
35	30	684147.00	Personal Loan	13.33	36	Approved
36	21	519296.00	Car Loan	8.23	24	Closed
37	71	410171.00	Personal Loan	7.37	24	Defaulted
38	76	480068.00	Car Loan	14.95	24	Approved
39	19	700411.00	Car Loan	14.29	12	Rejected
40	93	742649.00	Car Loan	13.12	24	Approved

TABLE VIEW

```
mysql> Select *from customers;
+-------------+--------------+------+--------+----------+-------------------+-----------+
| customer_id | name         | age  | gender | income   | employment_status | city      |
+-------------+--------------+------+--------+----------+-------------------+-----------+
|           1 | Customer_1   |   25 | Male   | 35000.00 | Salaried          | Delhi     |
|           2 | Customer_2   |   32 | Female | 48000.00 | Self-Employed     | Mumbai    |
|           3 | Customer_3   |   45 | Male   | 72000.00 | Business          | Bangalore |
|           4 | Customer_4   |   29 | Female | 29000.00 | Salaried          | Lucknow   |
|           5 | Customer_5   |   38 | Male   | 85000.00 | Business          | Hyderabad |
|           6 | Customer_6   |   41 | Female | 54000.00 | Salaried          | Pune      |
|           7 | Customer_7   |   27 | Male   | 31000.00 | Salaried          | Chennai   |
|           8 | Customer_8   |   50 | Female | 90000.00 | Business          | Kolkata   |
|           9 | Customer_9   |   34 | Male   | 60000.00 | Self-Employed     | Jaipur    |
|          10 | Customer_10  |   28 | Female | 45000.00 | Salaried          | Delhi     |
|          11 | Customer_11  |   39 | Male   | 78000.00 | Business          | Mumbai    |
|          12 | Customer_12  |   31 | Female | 52000.00 | Salaried          | Lucknow   |
|          13 | Customer_13  |   44 | Male   | 66000.00 | Self-Employed     | Pune      |
|          14 | Customer_14  |   26 | Female | 33000.00 | Salaried          | Chennai   |
|          15 | Customer_15  |   36 | Male   | 58000.00 | Business          | Hyderabad |
|          16 | Customer_16  |   48 | Female | 88000.00 | Business          | Delhi     |
|          17 | Customer_17  |   30 | Male   | 47000.00 | Salaried          | Bangalore |
|          18 | Customer_18  |   42 | Female | 69000.00 | Self-Employed     | Kolkata   |
|          19 | Customer_19  |   37 | Male   | 61000.00 | Salaried          | Mumbai    |
|          20 | Customer_20  |   29 | Female | 42000.00 | Salaried          | Lucknow   |
|          21 | Customer_21  |   53 | Male   | 87685.00 | Business          | Bangalore |
|          22 | Customer_22  |   35 | Female | 49843.00 | Salaried          | Delhi     |
|          23 | Customer_23  |   29 | Female | 60868.00 | Self-Employed     | Hyderabad |
|          24 | Customer_24  |   48 | Female | 93852.00 | Self-Employed     | Bangalore |
|          25 | Customer_25  |   52 | Female | 91439.00 | Salaried          | Lucknow   |
|          26 | Customer_26  |   40 | Female | 67685.00 | Self-Employed     | Delhi     |
|          27 | Customer_27  |   34 | Female | 41726.00 | Business          | Bangalore |
|          28 | Customer_28  |   39 | Female | 73817.00 | Self-Employed     | Hyderabad |
|          29 | Customer_29  |   53 | Male   | 91813.00 | Salaried          | Lucknow   |
|          30 | Customer_30  |   50 | Male   | 86453.00 | Salaried          | Delhi     |
|          31 | Customer_31  |   48 | Male   | 33723.00 | Self-Employed     | Delhi     |
|          32 | Customer_32  |   52 | Female | 58782.00 | Business          | Mumbai    |
|          33 | Customer_33  |   48 | Female | 33730.00 | Salaried          | Mumbai    |
|          34 | Customer_34  |   25 | Female | 43616.00 | Self-Employed     | Pune      |
|          35 | Customer_35  |   51 | Female | 88042.00 | Salaried          | Hyderabad |
|          36 | Customer_36  |   25 | Female | 63889.00 | Salaried          | Pune      |
|          37 | Customer_37  |   53 | Female | 67847.00 | Business          | Bangalore |
|          38 | Customer_38  |   43 | Male   | 75272.00 | Self-Employed     | Mumbai    |
|          39 | Customer_39  |   43 | Female | 90531.00 | Self-Employed     | Hyderabad |
|          40 | Customer_40  |   28 | Male   | 93202.00 | Business          | Mumbai    |
|          41 | Customer_41  |   38 | Female | 99742.00 | Business          | Hyderabad |
```

```
mysql> select *from Credit_History;
```

credit_id	customer_id	credit_score	past_defaults	credit_utilization
1	1	839	0	69.47
2	2	512	1	79.33
3	3	548	2	70.76
4	4	668	1	28.68
5	5	835	1	44.24
6	6	605	2	44.98
7	7	478	0	34.20
8	8	516	0	61.49
9	9	574	2	85.62
10	10	596	0	89.09
11	11	803	1	65.47
12	12	796	1	42.34
13	13	627	0	89.51
14	14	517	2	77.30
15	15	649	0	67.72
16	16	571	1	50.42
17	17	756	1	44.83
18	18	521	2	64.80
19	19	726	1	63.97
20	20	657	2	88.23
21	21	752	2	21.19
22	22	655	1	23.28
23	23	722	0	40.46
24	24	713	1	27.89
25	25	573	0	28.84
26	26	849	1	23.83
27	27	628	0	88.21
28	28	724	1	52.91
29	29	783	2	41.25
30	30	543	0	61.40
31	31	522	0	28.03
32	32	521	1	34.85
33	33	614	1	81.10
34	34	489	2	26.99
35	35	794	0	49.63
36	36	495	0	31.39
37	37	814	0	61.18
38	38	750	2	67.29
39	39	618	0	31.12
40	40	665	0	52.51

1-Overall Loan Default Rate

2-Average Loan Amount by Loan Type

3-Top 5 Highest Income Customers

```
mysql> select *from Payments;
Empty set (0.00 sec)

mysql> SELECT
    ->     ROUND(
    ->         COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0
    ->         / COUNT(*), 2
    ->     ) AS default_rate_percentage
    -> FROM Loans;
+-------------------------+
| default_rate_percentage |
+-------------------------+
|                   28.00 |
+-------------------------+
1 row in set (0.01 sec)

mysql> SELECT
    ->     loan_type,
    ->     ROUND(AVG(loan_amount),2) AS avg_loan_amount
    -> FROM Loans
    -> GROUP BY loan_type
    -> ORDER BY avg_loan_amount DESC;
+---------------+-----------------+
| loan_type     | avg_loan_amount |
+---------------+-----------------+
| Car Loan      |       580542.78 |
| Personal Loan |       579528.80 |
| Home Loan     |       525506.65 |
+---------------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT customer_id, name, income
    -> FROM Customers
    -> ORDER BY income DESC
    -> LIMIT 5;
+-------------+--------------+----------+
| customer_id | name         | income   |
+-------------+--------------+----------+
|          41 | Customer_41  | 99742.00 |
|         100 | Customer_100 | 99303.00 |
|          95 | Customer_95  | 98683.00 |
|          78 | Customer_78  | 97596.00 |
|          86 | Customer_86  | 97130.00 |
+-------------+--------------+----------+
```

1- Average Credit Score by Loan Status
2-Total Approved Loan Amount per City

```
mysql> SELECT
    ->     l.loan_status,
    ->     ROUND(AVG(ch.credit_score),2) AS avg_credit_score
    -> FROM Loans l
    -> JOIN Credit_History ch
    -> ON l.customer_id = ch.customer_id
    -> GROUP BY l.loan_status
    -> ORDER BY avg_credit_score DESC;
+-------------+------------------+
| loan_status | avg_credit_score |
+-------------+------------------+
| Defaulted   |           659.71 |
| Closed      |           643.50 |
| Approved    |           628.22 |
| Rejected    |           616.13 |
+-------------+------------------+
4 rows in set (0.01 sec)

mysql> SELECT
    ->     c.city,
    ->     ROUND(SUM(l.loan_amount),2) AS total_approved_amount
    -> FROM Customers c
    -> JOIN Loans l
    -> ON c.customer_id = l.customer_id
    -> WHERE l.loan_status = 'Approved'
    -> GROUP BY c.city
    -> ORDER BY total_approved_amount DESC;
+-----------+-----------------------+
| city      | total_approved_amount |
+-----------+-----------------------+
| Pune      |            2172073.00 |
| Delhi     |            1944094.00 |
| Mumbai    |            1791721.00 |
| Bangalore |            1390606.00 |
| Hyderabad |            1295153.00 |
| Lucknow   |            1117266.00 |
| Kolkata   |             866661.00 |
+-----------+-----------------------+
7 rows in set (0.00 sec)
```

High-Risk Customers

```
mysql> SELECT
    ->     c.customer_id,
    ->     c.name,
    ->     ch.credit_score,
    ->     ch.past_defaults
    -> FROM Customers c
    -> JOIN Credit_History ch
    -> ON c.customer_id = ch.customer_id
    -> WHERE ch.credit_score < 600
    ->    OR ch.past_defaults > 0
    -> ORDER BY ch.credit_score ASC;
+-------------+-------------+--------------+---------------+
| customer_id | name        | credit_score | past_defaults |
+-------------+-------------+--------------+---------------+
|          43 | Customer_43 |          453 |             2 |
|          69 | Customer_69 |          455 |             2 |
|          67 | Customer_67 |          459 |             1 |
|          56 | Customer_56 |          461 |             1 |
|          89 | Customer_89 |          471 |             0 |
|           7 | Customer_7  |          478 |             0 |
|          34 | Customer_34 |          489 |             2 |
|          62 | Customer_62 |          494 |             1 |
|          36 | Customer_36 |          495 |             0 |
|          92 | Customer_92 |          501 |             2 |
|          84 | Customer_84 |          507 |             1 |
|          94 | Customer_94 |          509 |             0 |
|           2 | Customer_2  |          512 |             1 |
|          98 | Customer_98 |          513 |             0 |
|           8 | Customer_8  |          516 |             0 |
|          59 | Customer_59 |          516 |             2 |
|          14 | Customer_14 |          517 |             2 |
|          18 | Customer_18 |          521 |             2 |
|          32 | Customer_32 |          521 |             1 |
|          31 | Customer_31 |          522 |             0 |
|          72 | Customer_72 |          531 |             2 |
|          74 | Customer_74 |          531 |             0 |
|          91 | Customer_91 |          534 |             2 |
|          57 | Customer_57 |          541 |             0 |
|          30 | Customer_30 |          543 |             0 |
|          80 | Customer_80 |          546 |             1 |
|           3 | Customer_3  |          548 |             2 |
|          82 | Customer_82 |          556 |             0 |
|          88 | Customer_88 |          561 |             2 |
|          77 | Customer_77 |          563 |             0 |
|          49 | Customer_49 |          565 |             1 |
```